



Guidance on the Measurement and Estimation of Height and Weight

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Measuring Weight

Principles

1. Ensure scales are calibrated regularly as per the protocol for the maintenance/calibration of measuring equipment in the MUST folder.
2. Place scales flat on level ground. Apply the brake for a wheelchair or ramp scale. Ensure the scale is at zero before commencing.
3. Ask the patient to remove heavy clothes or dressing gowns and empty pockets. Preferably, shoes should be removed. Specify if shoes were kept on or removed in the patient record.
4. Take the reading when it has stabilised.
5. Record the weight in kilograms and to the nearest 0.1kg.
6. All patient contact equipment should be cleaned as per manufacturers' guidance.
7. Contact the dietitian for advice if you are unable to obtain a weight.

Equipment required - the appropriate scales from the list below based on patients needs

Standing Scales/Flat scales:

1. Ensure the subject's feet are square on the scales.
2. Ensure the subject is standing freely e.g. not leaning on anyone or a zimmer.

Sitting Scales/Chair scales:

1. Ask the patient to sit upright with their back against the back of the chair.
2. Feet are to be off the ground and their hands on their lap.

Wheelchair Scales (for areas that have these):

1. Ensure the scale ramp(s) are parallel with their fronts aligned where appropriate e.g. Marsden ramps.
2. Weigh the wheelchair on its own. A record of the wheelchair weight should be made. Be sure to make a note of any cushions. Ensure the basket is empty.
3. Position the patient in the chair, and weigh with the chair.
4. Ensure feet are off the ground.
5. Subtract the chair weight from total weight to obtain the patients weight.

Full Body Hoist with Scale Attachment:

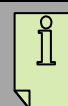
1. Place the scales on a hard, flat surface.
2. Secure the patient in the sling attached to the hoist.
3. Ensure the scale is switched on and at zero before lifting the patient.

Estimating weight in the presence of oedema or ascites

If the patient has signs of fluid overload from oedema or ascites use the table below to correct the weight before completing the Malnutrition Universal Screening Tool.

	Ascites	Peripheral Oedema
Minimal	2.2kg	1.0kg
Moderate	6.0kg	5.0kg
Severe	14.0kg	10.0kg

Example:

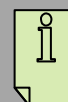


A patient has oedema up to the knees. He weighs 72kg. To estimate his dry weight, 5kg should be subtracted from his weight for moderate peripheral oedema. His estimated dry weight would be 67kg. The estimated dry weight should be used for the 'Malnutrition Universal Screening Tool' (MUST) calculations.

Estimating weight following amputation

Amputation	Add to the weight	Amputation	Add to the weight
Full leg	15.6%	Full arm	4.9%
Lower leg	4.5%	Forearm	1.6%
Foot	1.4%	Hand	0.6%

Example:



A patient has had their full arm amputated and weighs 54kg. To correct the weight 4.9% of this weight should be added.

$$(4.9/100) \times 54 = 2.6\text{Kg}$$

This equates to 2.6kg, meaning the corrected weight is 56.6kg (54kg + 2.6kg). The corrected weight should be used for 'Malnutrition Universal Screening Tool' (MUST) calculations.

Estimating weight with a plaster cast

Advice should be sought from the dietitian to determine how much the plaster cast weighs and the corrected weight.

Measuring height using a height stick

Equipment required: Height stick

Ensure the height stick is vertical.

Remove heavy clothes, hats and shoes (if safe to do so).

Ask the patient to stand straight and look straight ahead. Feet should be flat and heels against the height stick.

The head plate of the height stick should then be gently lowered until it touches the crown of the subject's head.

Record the measurement and whether the patient had shoes on or not.

Estimating height using ulna length (Not appropriate for Learning Disabilities)

Equipment required: Measuring Tape

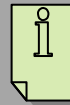
1. **Remove any watches or jewellery from the wrist area.**
2. **Ask the patient to bend their arm (left preferably), palm across chest, with their fingers pointing to the opposite shoulder.**
3. **Measure the length between the point of the elbow and the mid-point of the prominent bone of the wrist as indicated by the marks on the diagram.**



4. **Record the reading to the nearest 0.5cm.**
5. **Measuring tape, if disposable should be disposed of after single patient use and if re-usable should be cleaned with detergent wipes and dried.**

6. Use Appendix (i) to estimate height from ulna length.

Example:



A 48 year old lady has an ulna length of 26cm. This would make her estimated height as 1.68m. (An extract of Appendix (i) is detailed below)

Men (< 65 years) Men (> 65 years)	1.73 1.68
Ulna length (cm)	26
Women (< 65 years) Women (> 65 years)	1.68 1.65

**Estimating height using knee height
(Not appropriate for Learning Disabilities)**

General recommendations:

This measurement can be either taken with the patient lying down or in a sitting position. Either way a tape measure or knee height calliper can be used for this measurement. Footwear should be removed.

Equipment required: Knee height calliper/Tape Measure

a. Knee height calliper

- The patient should bend their left knee and left ankle to a right angle
- Place the fixed blade under the patients heel
- Press the sliding blade down against the thigh about 4cm behind their knee cap. The shaft of the calliper should be in line with the long bone in the lower leg and over the ankle bone. Measure to the nearest 0.5cm.
- Clean the calliper after each use with a detergent wipe and store dry.
- See Appendix (ii) for conversion to height

b. Tape measure

- The patient should bend their left knee and left ankle to a right angle
- Hold the tape measure between 3rd and 4th fingers with a zero reading underneath fingers. Place your hand about 4cm behind the front of the knee
- Extend the tape measure straight down the side of the leg in line with the bony prominence at the ankle
- Measure to the nearest 0.5cm

- **Measuring tape, if disposable should be disposed of after single patient use and if re-usable should be cleaned with detergent wipes and dried.**
- See Appendix (ii) for conversion to height

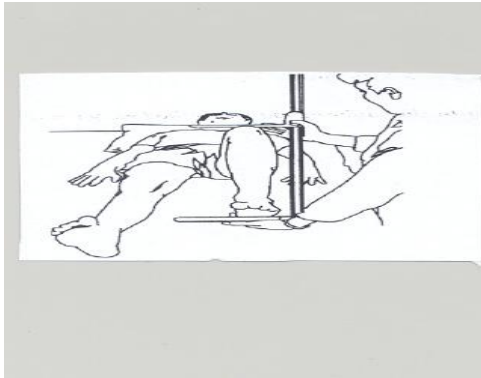


Illustration 1:
Using the knee height calliper

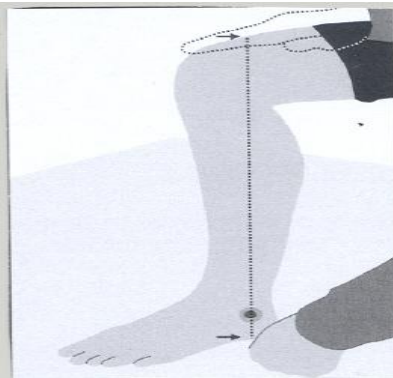
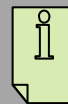


Illustration 2:
Using the tape measure

Example:



A 65 year old male has had his knee height measured at 57cm. This would estimate his height 1.78m. (An extract of Appendix (ii) is detailed below)

Men (18 – 59 years)	1.80	1.79
Men (60 – 90 years)	1.79	1.78
Knee height (cm)	57.5	57cm
Women (18 – 59 years)	1.75	1.74

Estimating height in Learning Disabilities

If height can not be measured:

Use self reported height - use recently documented or self-reported height (if reliable and realistic).

Use a tape measure

Equipment required: Measuring Tape.

Technique:

Client should be lying down on their bed or a floor mat on their back (supine). Measure the length of the body in small sections to provide a total length.

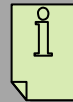
1. Place clip board on top of patients' head, measure straight down from mid point of head to a point on the spine that is in line with the top of the shoulder. Measure from top of shoulder to top of hip bone, then from this point to back of knee and finally back of knee to the back of the heel.
2. Record each individual measurement in cm and add together. This will give you an estimated height.
3. If the patient has marked scoliosis, measure both sides using the above technique, add the figures together and divide by 2 to give you an estimated height.
4. Height should be recorded to the nearest 0.5cm.
5. Measuring tape, if disposable should be disposed of after single use and if re-usable should be cleaned with detergent wipes and dried.

STEP 1 of the 'Malnutrition Universal Screening Tool' ('MUST'): Calculating Body Mass Index

Body Mass Index (BMI) can be calculated using either of the techniques below:

1. Using the Step 1 ready reckoner
2. Using the formula $BMI = \frac{\text{Weight (kg)}}{\text{Height (m)}^2}$

Example using the equation:



A patient weighs 64kg and is 1.75m.

$$BMI = \frac{64\text{kg}}{1.75\text{m} \times 1.75} = 20.9$$

The BMI is 20.9kg/m². This means they are of desirable weight and the Step 1 score is 0.

Interpretation of BMI

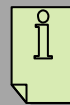
BMI	Interpretation	MUST Score
< 18.5	Underweight	2
18.5 – 20	Underweight	1
20-25	Desirable weight	0
25-30	Overweight	0
> 30	Obese	0

STEP 2 of the 'Malnutrition Universal Screening Tool' ('MUST'): Calculating Percentage Weight Loss

Percentage weight loss can be calculated using either of the techniques below. To calculate this accurately, ask the patient their weight 3-6 months ago and not the week to week weight.

1. Using the ready reckoner
2. % weight loss =
$$\frac{\text{weight 3 – 6 months ago (kg)} - \text{current weight (kg)}}{\text{weight 3 – 6 months ago (kg)}} \times 100$$

Example using the equation:



The patient currently weighs 71kg but weighed 80kg 4 months ago.

$$\frac{80\text{kg} - 71\text{kg}}{80\text{kg}} \times 100 = 11.3\%$$

The patient has lost 11% of their body weight in 4 months. This means their weight loss is clinically significant and the Step 2 score is 2.

Interpretation of Weight Loss

% weight loss	Interpretation	MUST Score
> 10%	Clinically significant	2
5-10%	More than normal variation, increased risk of malnutrition	1
< 5%	Within normal variation	0

References

British Association of Parenteral & Enteral Nutrition. (2003) The 'MUST' Explanatory Booklet: BAPEN

The Parenteral and Enteral Nutrition Group of the British Dietetic Association (2007) A Pocket Guide to Clinical Nutrition: PEN Group

Appendix i: Calculating height from ulna length

Height (m)														
Men (< 65 years)	1.94	1.93	1.91	1.89	1.87	1.85	1.84	1.82	1.80	1.78	1.76	1.75	1.73	1.71
Men (> 65 years)	1.87	1.86	1.84	1.82	1.81	1.79	1.78	1.76	1.75	1.73	1.71	1.70	1.68	1.67
Ulna length (cm)	32.0	31.5	31.0	30.5	30.0	29.5	29.0	28.5	28.0	27.5	27.0	26.5	26.0	25.5
Height (m)														
Women (< 65 years)	1.84	1.83	1.81	1.80	1.79	1.77	1.76	1.75	1.73	1.72	1.70	1.69	1.68	1.66
Women (> 65 years)	1.84	1.83	1.81	1.79	1.78	1.76	1.75	1.73	1.71	1.70	1.68	1.66	1.65	1.63

Height (m)														
Men (< 65 years)	1.69	1.67	1.66	1.64	1.62	1.60	1.58	1.57	1.55	1.53	1.51	1.49	1.48	1.46
Men (> 65years)	1.65	1.63	1.62	1.60	1.59	1.57	1.56	1.54	1.52	1.51	1.49	1.48	1.46	1.45
Ulna length (cm)	25.0	24.5	24.0	23.5	23.0	22.5	22.0	21.5	21.0	20.5	20.0	19.5	19.0	18.5
Height (m)														
Women (< 65 years)	1.65	1.63	1.62	1.61	1.59	1.58	1.56	1.55	1.54	1.52	1.51	1.50	1.48	1.47
Women (> 65 years)	1.61	1.60	1.58	1.56	1.55	1.53	1.52	1.50	1.48	1.47	1.45	1.44	1.42	1.40

Appendix ii: Calculating height from knee height

Height (m)															
Men (18 – 59 years)	1.94	1.93	1.92	1.91	1.90	1.89	1.88	1.87	1.865	1.86	1.85	1.84	1.83	1.82	1.81
Men (60 – 90 years)	1.94	1.93	1.92	1.91	1.90	1.89	1.88	1.87	1.86	1.85	1.84	1.83	1.82	1.81	1.80
Knee height (cm)	65	64.5	64	63.5	63	62.5	62	61.5	61	60.5	60	59.5	59	58.5	58
Height (m)															
Women (18 – 59 years)	1.89	1.88	1.875	1.87	1.86	1.85	1.84	1.83	1.82	1.81	1.80	1.79	1.78	1.77	1.76
Women (60 – 90 years)	1.86	1.85	1.84	1.835	1.83	1.82	1.81	1.80	1.79	1.78	1.77	1.76	1.75	1.74	1.73

Height (m)															
Men (18 – 59 years)	1.80	1.79	1.78	1.77	1.76	1.75	1.74	1.73	1.72	1.71	1.705	1.70	1.69	1.68	1.67
Men (60 – 90 years)	1.79	1.78	1.77	1.76	1.74	1.73	1.72	1.71	1.70	1.69	1.68	1.67	1.66	1.65	1.64
Knee height (cm)	57.5	57	56.5	56	55.5	55	54.5	54	53.5	53	52.5	52	51.5	51	50.5
Height (m)															
Women (18 – 59 years)	1.75	1.74	1.735	1.73	1.72	1.71	1.70	1.69	1.68	1.67	1.66	1.65	1.64	1.63	1.62
Women (60 – 90 years)	1.72	1.71	1.70	1.69	1.68	1.67	1.66	1.65	1.64	1.63	1.625	1.62	1.61	1.60	1.59

Height (m)															
Men (18 – 59 years)	1.66	1.65	1.64	1.63	1.62	1.61	1.60	1.59	1.58	1.57	1.56	1.555	1.55	1.54	1.53
Men (60 – 90 years)	1.63	1.62	1.61	1.60	1.59	1.58	1.57	1.56	1.55	1.54	1.53	1.52	1.51	1.49	1.48
Knee height (cm)	50	49.5	49	48.5	48	47.5	47	46.5	46	45.5	45	44.5	44	43.5	43
Height (m)															
Women (18 – 59 years)	1.61	1.60	1.59	1.585	1.58	1.57	1.56	1.55	1.54	1.53	1.52	1.51	1.50	1.49	1.48
Women (60 – 90 years)	1.58	1.57	1.56	1.55	1.54	1.53	1.52	1.51	1.50	1.49	1.48	1.47	1.46	1.45	1.44